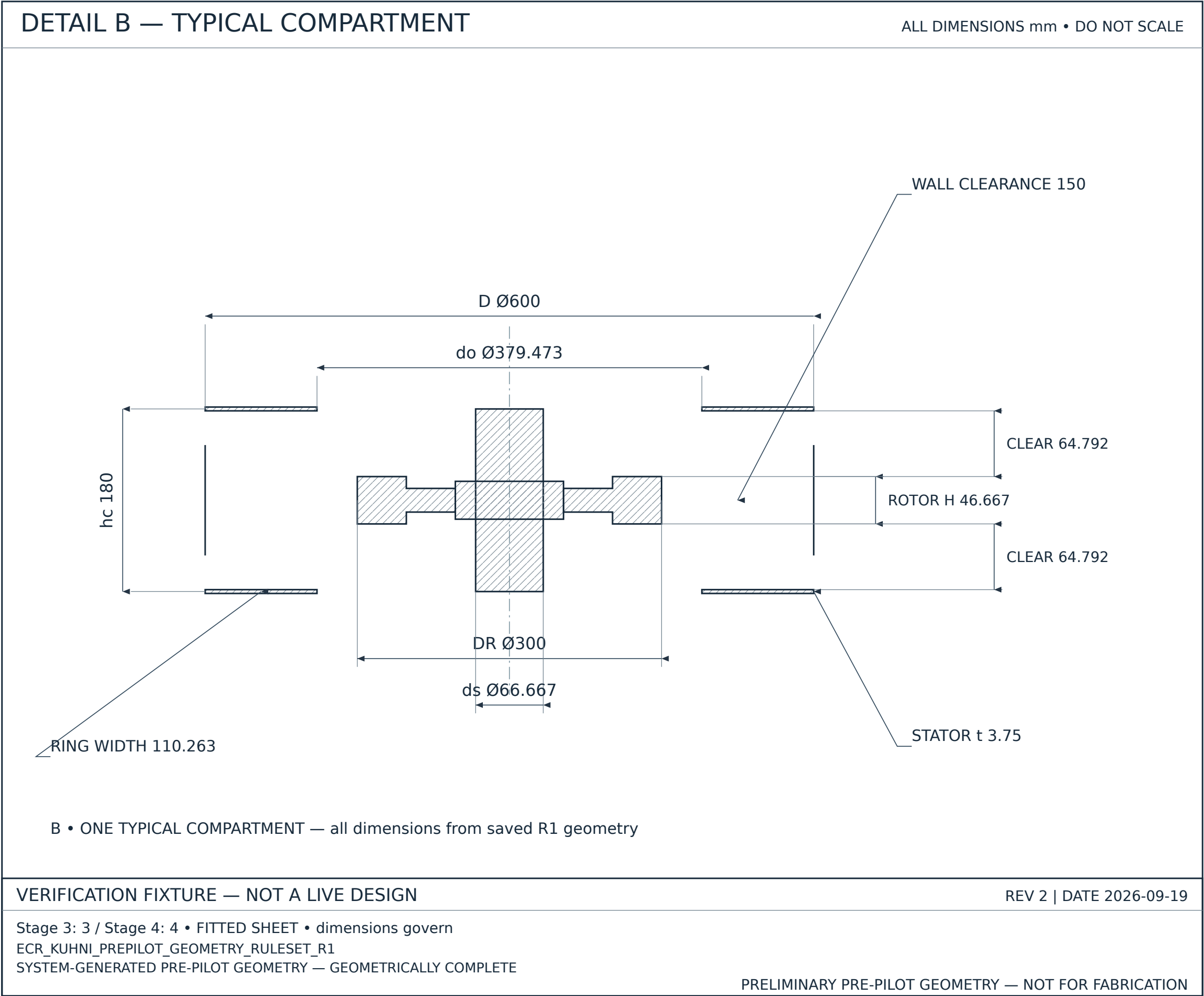
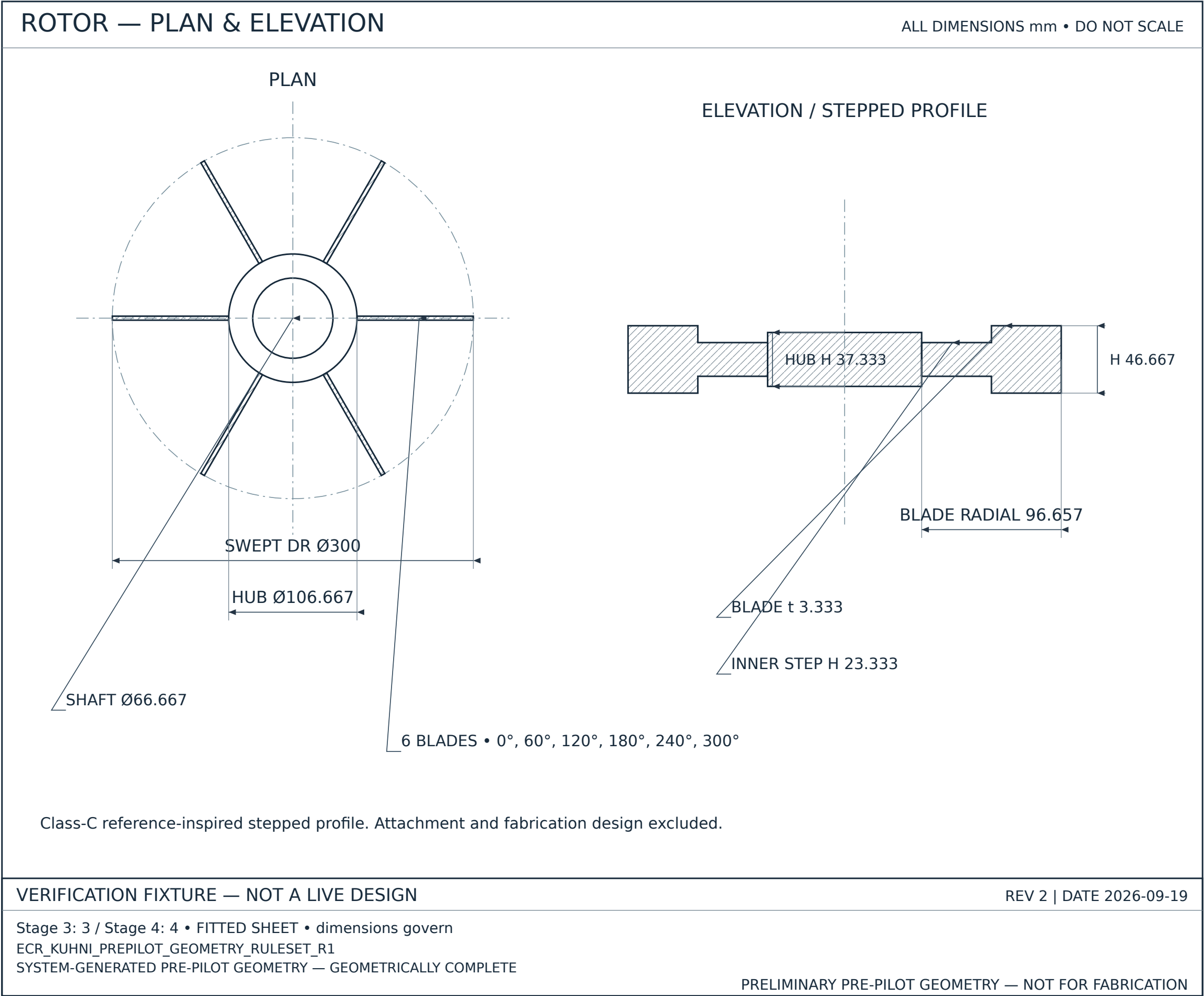
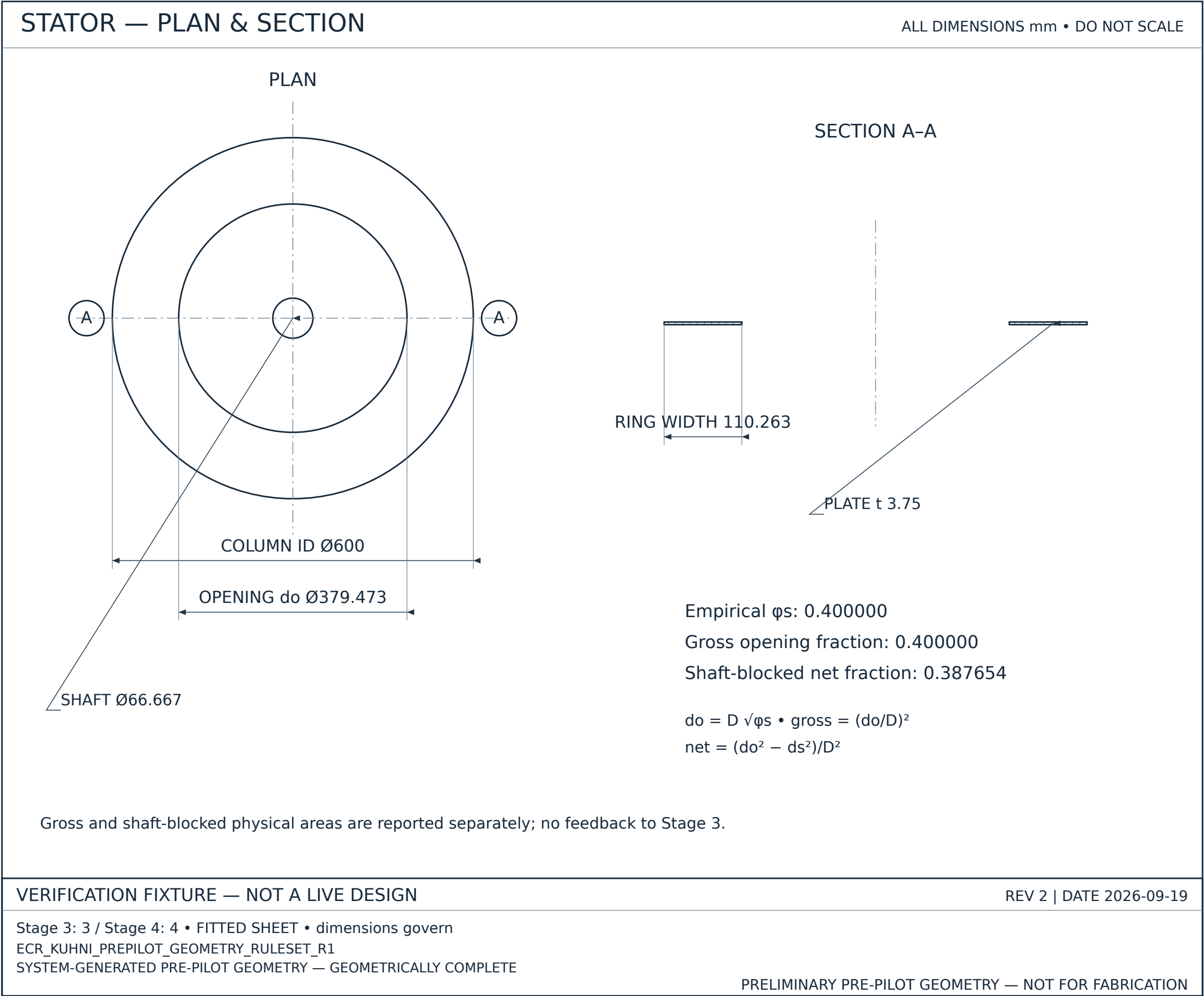








Geometry summary — frozen revision dimensions

PRELIMINARY PRE-PILOT GEOMETRY — NOT FOR FABRICATION

Geometrically complete pre-pilot layout only; not fabrication or mechanical qualification.

stage3ResultId: 3 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

stage4ResultId: 4 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

columnDiameterM: 0.6 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rotorDiameterM: 0.3 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rotorDiameterRatio: 0.5 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

compartmentHeightM: 0.18 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

compartmentCount: 39 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

requiredActiveHeightM: 7 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

installedActiveHeightM: 7.02 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

designNt: 7 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

hetsM: 1 m | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

statorFreeAreaRatio: 0.4 | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

selectedRpm: 50 rpm | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rpmMin: 30 rpm | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

rpmMax: 70 rpm | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

phaseConfiguration: NMP continuous / RRBO dispersed | Inherited | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

shaftDiameterM: 0.0666666666666667 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. (10/45) DR; reference-inspired scaling, not strength qualification.

statorThicknessM: 0.00375 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. min(D/160,hc/20); geometric plate envelope.

bladeCount: 6 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class B. Garthe reference construction family, Table A.1.

bladeHeightM: 0.0466666666666667 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. min((7/45) DR,0.60(hc-ts)); reference-inspired clearance-capped envelope.

hubDiameterM: 0.1066666666666667 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 1.60 ds; attachment envelope.

hubHeightM: 0.03733333333333336 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 0.80 HR; within rotor envelope.

bladeThicknessM: 0.00333333333333333 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. min(DR/90,HR/10,dh/12); envelope only.

bladeRadialLengthM: 0.09665740712161014 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. (sqrt(DR²-tb²)-dh)/2; exact outer-corner swept diameter.

bladeInnerSegmentLengthM: 0.04832870356080507 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Lb/2; approved R1 stepped approximation, not exact reference reproduction.

bladeOuterSegmentLengthM: 0.04832870356080507 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Lb/2; approved R1 stepped approximation.

bladeInnerSegmentHeightM: 0.02333333333333334 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. HR/2, centred about rotor midplane.

bladeOuterSegmentHeightM: 0.0466666666666667 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. HR, centred about rotor midplane.

rotorOffsetM: 0.09 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. hc/2; centred rotor.

bottomDisengagementM: 0.6 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D; packaging allocation, not separation-time qualification.

topDisengagementM: 0.6 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D; packaging allocation, not separation-time qualification.

bottomHeadDepthM: 0.15 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D/4; 2:1 ellipsoidal internal envelope.

topHeadDepthM: 0.15 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D/4; 2:1 ellipsoidal internal envelope.

headStraightFlangeM: 0 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Zero in R1 internal envelope model.

supportHeightM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D/2; skirt below vessel datum.

skirtDiameterM: 0.6 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D; centred cylindrical envelope.

skirtAccessWidthM: 0.18 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 0.30D; access opening at azimuth zero.

skirtAccessHeightM: 0.15 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 0.25D; access opening.

skirtAccessElevationM: -0.15 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. -0.25D; centre elevation.

lowerShaftSupportM: 0.4499999999999996 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. hB+LB/2.

upperShaftSupportM: 8.07 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. z1+LT/2.

supportArmWidthM: 0.012 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 0.02D; tangential width, three arms at 30/150/270 degrees.

supportArmDepthM: 0.024 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 0.04D; axial depth.

supportHousingDiameterM: 0.1 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 1.5ds; central envelope.

supportHousingHeightM: 0.024 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 0.04D; central envelope.

supportArmRadialStartM: 0.05 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 0.75ds; housing surface.

supportArmRadialEndM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D/2; shell interface.

shaftLowerTerminationM: 0.4499999999999996 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Lower support centre.

shaftUpperTerminationM: 8.67 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. HV+D/4; drive engagement.

shaftLengthM: 8.22 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Upper minus lower termination.

sealElevationM: 8.52 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Top pole; coaxial with shaft.

sealDiameterM: 0.13333333333333333 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 2ds; reserved housing, not product selection.

sealHeightM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 0.10D above top pole.

drivePedestalDiameterM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D/2; coaxial reserved envelope.

drivePedestalHeightM: 0.15 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D/4 above top pole.

driveBodyDiameterM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D/2; reserved envelope.

driveBodyHeightM: 0.44999999999999996 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. 3D/4 above pedestal.

driveHeightM: 0.6 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D including pedestal; not gearbox adequacy.

statorOpeningDiameterM: 0.3794733192202055 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. D sqrt(phi_s); gross opening convention, no shaft compensation.

columnAreaM2: 0.2827433388230814 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. pi D²/4.

shaftAreaM2: 0.0034906585039886587 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. pi ds²/4.

statorGrossOpenAreaM2: 0.11309733552923253 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. pi do²/4.

statorNetOpenAreaM2: 0.10960667702524389 m² | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. pi(do²-ds²)/4; no upstream feedback.

grossFreeAreaRatio: 0.4 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. (do/D)² = phi_s.

shaftBlockedFreeAreaRatio: 0.3876543209876543 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. (do²-ds²)/D²; separate physical report.

calculatedFreeAreaRatio: 0.4 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Gross ratio, not legacy net-area convention.

rotorWallClearanceM: 0.15 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. (D-DR)/2.

shaftOpeningClearanceM: 0.15640332627676942 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. (do-ds)/2.

statorRadialWidthM: 0.11026334038989724 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. (D-do)/2.

rotorAxialEnvelopeM: 0.04666666666666667 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Full outer blade axial height.

rotorLowerClearanceM: 0.06479166666666666 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. (hc-ts-HR)/2.

rotorUpperClearanceM: 0.06479166666666666 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. (hc-ts-HR)/2.

activeStartM: 0.75 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. hB+LB.

activeEndM: 7.77 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. z0+HA.

topTangentM: 8.37 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. z1+LT.

vesselHeightM: 8.52 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. HA+2.5D.

overallHeightM: 9.42 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Support+vessel+drive = HA+4D.

rotorThicknessM: Not applicable — stepped rotor, not flat disc | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Explicit N/A, no missing construction dimension.

rotorConstruction: R1 six-blade stepped rotor | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Reference-inspired family; explicitly approved engineering approximation.

statorConstruction: Annular single central opening | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class B. Garthe / Weber-Jupke reference family.

flowArrangement: nmp-down-rrbo-up | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Explicit application layout; continuity and upstream hydraulics unchanged.

headProfiles: 2:1 ellipsoidal | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Internal envelope only.

bladeAzimuthsDeg: 0, 60, 120, 180, 240, 300 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Six equally spaced blades; same clocking in every compartment.

supportAzimuthsDeg: 30, 150, 270 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Approved three-arm support clocking.

skirtAccessAzimuthDeg: 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Approved skirt access orientation.

designEnvelope: hc/D 0.20–0.30; DR/D 0.33–0.50; phi_s 0.20–0.40 | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class B. Perry 8e p15-83 → Pratt-Stevens; advisory, never clamp inherited geometry.

P01-elevationM: 0.51 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-boreM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-azimuthDeg: 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-radialOffsetM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-outsideDiameterM: 0.072 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P01-projectionM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

P02-elevationM: 0.39 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-boreM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-azimuthDeg: 180 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-radialOffsetM: 0.3 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-outsideDiameterM: 0.072 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P02-projectionM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P03-elevationM: 8.01 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

P03-boreM: 0.06 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

[illegible]

[illegible]

compartment31-rotorM: 6.24 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment32-bottomM: 6.33 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment32-topM: 6.51 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment32-rotorM: 6.42 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment33-bottomM: 6.51 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment33-topM: 6.689999999999995 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment33-rotorM: 6.6 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment34-bottomM: 6.689999999999995 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment34-topM: 6.87 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment34-rotorM: 6.779999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment35-bottomM: 6.87 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment35-topM: 7.05 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment35-rotorM: 6.96 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment36-bottomM: 7.05 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment36-topM: 7.229999999999995 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment36-rotorM: 7.14 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment37-bottomM: 7.229999999999995 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment37-topM: 7.41 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment37-rotorM: 7.319999999999999 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment38-bottomM: 7.41 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment38-topM: 7.59 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment38-rotorM: 7.5 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment39-bottomM: 7.59 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment39-topM: 7.77 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

compartment39-rotorM: 7.68 m | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Compartment and internals schedules

- Compartment 1: bottom 0.75 m; top 0.9299999999999999 m; rotor 0.84 m; stator 0.9299999999999999 m
- Compartment 2: bottom 0.9299999999999999 m; top 1.1099999999999999 m; rotor 1.02 m; stator 1.1099999999999999 m
- Compartment 3: bottom 1.1099999999999999 m; top 1.29 m; rotor 1.2 m; stator 1.29 m
- Compartment 4: bottom 1.29 m; top 1.47 m; rotor 1.38 m; stator 1.47 m
- Compartment 5: bottom 1.47 m; top 1.65 m; rotor 1.56 m; stator 1.65 m
- Compartment 6: bottom 1.65 m; top 1.83 m; rotor 1.74 m; stator 1.83 m
- Compartment 7: bottom 1.83 m; top 2.01 m; rotor 1.92 m; stator 2.01 m
- Compartment 8: bottom 2.01 m; top 2.19 m; rotor 2.0999999999999996 m; stator 2.19 m
- Compartment 9: bottom 2.19 m; top 2.37 m; rotor 2.2800000000000002 m; stator 2.37 m
- Compartment 10: bottom 2.37 m; top 2.55 m; rotor 2.46 m; stator 2.55 m
- Compartment 11: bottom 2.55 m; top 2.73 m; rotor 2.6399999999999997 m; stator 2.73 m
- Compartment 12: bottom 2.73 m; top 2.91 m; rotor 2.82 m; stator 2.91 m
- Compartment 13: bottom 2.91 m; top 3.09 m; rotor 3 m; stator 3.09 m
- Compartment 14: bottom 3.09 m; top 3.27 m; rotor 3.1799999999999997 m; stator 3.27 m
- Compartment 15: bottom 3.27 m; top 3.4499999999999997 m; rotor 3.36 m; stator 3.4499999999999997 m
- Compartment 16: bottom 3.4499999999999997 m; top 3.63 m; rotor 3.54 m; stator 3.63 m
- Compartment 17: bottom 3.63 m; top 3.81 m; rotor 3.7199999999999998 m; stator 3.81 m
- Compartment 18: bottom 3.81 m; top 3.9899999999999998 m; rotor 3.9 m; stator 3.9899999999999998 m
- Compartment 19: bottom 3.9899999999999998 m; top 4.17 m; rotor 4.08 m; stator 4.17 m
- Compartment 20: bottom 4.17 m; top 4.35 m; rotor 4.26 m; stator 4.35 m
- Compartment 21: bottom 4.35 m; top 4.529999999999999 m; rotor 4.4399999999999995 m; stator 4.529999999999999 m
- Compartment 22: bottom 4.529999999999999 m; top 4.71 m; rotor 4.619999999999999 m; stator 4.71 m
- Compartment 23: bottom 4.71 m; top 4.89 m; rotor 4.8 m; stator 4.89 m
- Compartment 24: bottom 4.89 m; top 5.07 m; rotor 4.9799999999999995 m; stator 5.07 m
- Compartment 25: bottom 5.07 m; top 5.25 m; rotor 5.16 m; stator 5.25 m
- Compartment 26: bottom 5.25 m; top 5.43 m; rotor 5.34 m; stator 5.43 m
- Compartment 27: bottom 5.43 m; top 5.609999999999999 m; rotor 5.52 m; stator 5.609999999999999 m
- Compartment 28: bottom 5.609999999999999 m; top 5.79 m; rotor 5.7 m; stator 5.79 m
- Compartment 29: bottom 5.79 m; top 5.97 m; rotor 5.88 m; stator 5.97 m
- Compartment 30: bottom 5.97 m; top 6.1499999999999995 m; rotor 6.06 m; stator 6.1499999999999995 m
- Compartment 31: bottom 6.1499999999999995 m; top 6.33 m; rotor 6.24 m; stator 6.33 m
- Compartment 32: bottom 6.33 m; top 6.51 m; rotor 6.42 m; stator 6.51 m
- Compartment 33: bottom 6.51 m; top 6.6899999999999995 m; rotor 6.6 m; stator 6.6899999999999995 m
- Compartment 34: bottom 6.6899999999999995 m; top 6.87 m; rotor 6.779999999999999 m; stator 6.87 m
- Compartment 35: bottom 6.87 m; top 7.05 m; rotor 6.96 m; stator 7.05 m
- Compartment 36: bottom 7.05 m; top 7.2299999999999995 m; rotor 7.14 m; stator 7.2299999999999995 m
- Compartment 37: bottom 7.2299999999999995 m; top 7.41 m; rotor 7.319999999999999 m; stator 7.41 m
- Compartment 38: bottom 7.41 m; top 7.59 m; rotor 7.5 m; stator 7.59 m
- Compartment 39: bottom 7.59 m; top 7.77 m; rotor 7.68 m; stator 7.77 m

R | R1 six-blade stepped rotor — Class C reference-inspired approximation | count 39 | diameter 0.3 m | thickness 0.04666666666666667 m | elevations 0.84, 1.02, 1.2, 1.38, 1.56, 1.74, 1.92, 2.0999999999999996, 2.2800000000000002, 2.46, 2.6399999999999997, 2.82, 3, 3.1799999999999997, 3.36, 3.54, 3.7199999999999998, 3.9, 4.08, 4.26, 4.4399999999999995, 4.619999999999999, 4.8, 4.9799999999999995, 5.16, 5.34, 5.52, 5.7, 5.88, 6.06, 6.24, 6.42, 6.6, 6.779999999999999, 6.96, 7.14, 7.319999999999999, 7.5, 7.68 m

S | Annular ring stator — single gross central opening | count 40 | diameter 0.6 m | thickness 0.00375 m | elevations 0.75, 0.9299999999999999, 1.1099999999999999, 1.29, 1.47, 1.65, 1.83, 2.01, 2.19, 2.37, 2.55, 2.73, 2.91, 3.09, 3.27, 3.4499999999999997, 3.63, 3.81, 3.9899999999999998, 4.17, 4.35, 4.529999999999999, 4.71, 4.89, 5.07, 5.25, 5.43, 5.609999999999999, 5.79, 5.97, 6.1499999999999995, 6.33, 6.51, 6.6899999999999995, 6.87, 7.05, 7.2299999999999995, 7.41, 7.59, 7.77 m

SH | Continuous shaft envelope; mechanical qualification excluded | count 1 | diameter 0.06666666666666667 m | thickness 0.06666666666666667 m | elevations 0.44999999999999996, 8.67 m

SUP | Three-arm shaft supports, azimuths 30/150/270 deg | count 2 | diameter 0.6 m | thickness 0.024 m | elevations 0.44999999999999996, 8.07 m

Preliminary nozzle / connection schedule

- P01 | RRBO feed | bottom | elevation 0.51 m | bore 0.06 m | azimuth 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.072 m; projection 0.06 m; radial offset 0.3 m
- P02 | NMP-rich extract outlet | bottom | elevation 0.39 m | bore 0.06 m | azimuth 180 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.072 m; projection 0.06 m; radial offset 0.3 m
- P03 | NMP feed | top | elevation 8.01 m | bore 0.06 m | azimuth 180 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.072 m; projection 0.06 m; radial offset 0.3 m
- P04 | RRBO-rich raffinate outlet | top | elevation 8.129999999999999 m | bore 0.06 m | azimuth 0 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.072 m; projection 0.06 m; radial offset 0.3 m
- A01 | Low-shell flush | bottom | elevation 0.27 m | bore 0.024 m | azimuth 270 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.0288 m; projection 0.06 m; radial offset 0.3 m
- A02 | High-shell equalization | top | elevation 8.25 m | bore 0.024 m | azimuth 270 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.0288 m; projection 0.06 m; radial offset 0.3 m
- S01 | Lower sample connection | bottom | elevation 0.63 m | bore 0.015 m | azimuth 90 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.018 m; projection 0.06 m; radial offset 0.3 m
- S02 | Upper sample connection | top | elevation 7.89 m | bore 0.015 m | azimuth 90 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.018 m; projection 0.06 m; radial offset 0.3 m
- I01 | Lower instrument connection | bottom | elevation 0.32999999999999996 m | bore 0.015 m | azimuth 45 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.018 m; projection 0.06 m; radial offset 0.3 m
- I02 | Upper instrument connection | top | elevation 7.949999999999999 m | bore 0.015 m | azimuth 45 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing. | axis radial; OD 0.018 m; projection 0.06 m; radial offset 0.3 m
- D01 | Bottom-head drain | bottom-head | elevation 0 m | bore 0.024 m | azimuth N/A axial deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bottom-pole axial drain; azimuth N/A. | axis down; OD 0.0288 m; projection 0.06 m; radial offset 0 m
- V01 | Top-head crown-region vent | top-head | elevation 8.489999999999998 m | bore 0.024 m | azimuth 315 deg | System-generated | ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: r=0.30D, ellipsoidal surface z=zT+0.20D; not absolute-high-point or vent-duty qualification. | axis up; OD 0.0288 m; projection 0.06 m; radial offset 0.18 m

Validation, assumptions and unresolved items

PASS: Frozen sources must be identified, current and compatible.

PASS: columnDiameterM must be finite and positive.

PASS: rotorDiameterM must be finite and positive.

PASS: rotorDiameterRatio must be finite and positive.

PASS: compartmentHeightM must be finite and positive.

PASS: compartmentCount must be finite and positive.

PASS: installedActiveHeightM must be finite and positive.

PASS: requiredActiveHeightM must be finite and positive.

PASS: designNt must be finite and positive.

PASS: hetsM must be finite and positive.

PASS: selectedRpm must be finite and positive.

PASS: rpmMin must be finite and positive.

PASS: rpmMax must be finite and positive.

PASS: statorFreeAreaRatio must be finite and positive.

PASS: Positive integer count; drawing capacity 1000 compartments.

PASS: Frozen count × pitch must equal frozen installed height; no repair.

PASS: Frozen required height must agree with Nt × HETS and fit installed height.

PASS: Frozen rotor diameter/ratio must agree and fit column.

PASS: Frozen selected RPM must lie inside frozen window.

PASS: Frozen phase continuity is required.

PASS: Empirical free-area input must be between zero and one.

PASS: columnDiameterM must be finite.

PASS: rotorDiameterM must be finite.

PASS: rotorDiameterRatio must be finite.

PASS: compartmentHeightM must be finite.

PASS: compartmentCount must be finite.

PASS: requiredActiveHeightM must be finite.

PASS: installedActiveHeightM must be finite.

PASS: designNt must be finite.

PASS: hetsM must be finite.

PASS: statorFreeAreaRatio must be finite.

PASS: selectedRpm must be finite.

PASS: rpmMin must be finite.

PASS: rpmMax must be finite.

PASS: shaftDiameterM must be finite.

PASS: statorThicknessM must be finite.

PASS: bladeCount must be finite.

PASS: bladeHeightM must be finite.

PASS: hubDiameterM must be finite.

PASS: hubHeightM must be finite.

PASS: bladeThicknessM must be finite.

PASS: bladeRadialLengthM must be finite.

PASS: bladeInnerSegmentLengthM must be finite.

PASS: bladeOuterSegmentLengthM must be finite.

PASS: bladeInnerSegmentHeightM must be finite.

PASS: bladeOuterSegmentHeightM must be finite.

PASS: rotorOffsetM must be finite.

PASS: bottomDisengagementM must be finite.

PASS: topDisengagementM must be finite.

PASS: bottomHeadDepthM must be finite.

PASS: topHeadDepthM must be finite.
PASS: headStraightFlangeM must be finite.
PASS: supportHeightM must be finite.
PASS: skirtDiameterM must be finite.
PASS: skirtAccessWidthM must be finite.
PASS: skirtAccessHeightM must be finite.
PASS: skirtAccessElevationM must be finite.
PASS: lowerShaftSupportM must be finite.
PASS: upperShaftSupportM must be finite.
PASS: supportArmWidthM must be finite.
PASS: supportArmDepthM must be finite.
PASS: supportHousingDiameterM must be finite.
PASS: supportHousingHeightM must be finite.
PASS: supportArmRadialStartM must be finite.
PASS: supportArmRadialEndM must be finite.
PASS: shaftLowerTerminationM must be finite.
PASS: shaftUpperTerminationM must be finite.
PASS: shaftLengthM must be finite.
PASS: sealElevationM must be finite.
PASS: sealDiameterM must be finite.
PASS: sealHeightM must be finite.
PASS: drivePedestalDiameterM must be finite.
PASS: drivePedestalHeightM must be finite.
PASS: driveBodyDiameterM must be finite.
PASS: driveBodyHeightM must be finite.
PASS: driveHeightM must be finite.
PASS: statorOpeningDiameterM must be finite.
PASS: columnAreaM2 must be finite.
PASS: shaftAreaM2 must be finite.
PASS: statorGrossOpenAreaM2 must be finite.
PASS: statorNetOpenAreaM2 must be finite.
PASS: grossFreeAreaRatio must be finite.
PASS: shaftBlockedFreeAreaRatio must be finite.
PASS: calculatedFreeAreaRatio must be finite.
PASS: rotorWallClearanceM must be finite.
PASS: shaftOpeningClearanceM must be finite.
PASS: statorRadialWidthM must be finite.
PASS: rotorAxialEnvelopeM must be finite.
PASS: rotorLowerClearanceM must be finite.
PASS: rotorUpperClearanceM must be finite.
PASS: activeStartM must be finite.
PASS: activeEndM must be finite.
PASS: topTangentM must be finite.
PASS: vesselHeightM must be finite.
PASS: overallHeightM must be finite.
PASS: skirtAccessAzimuthDeg must be finite.
PASS: P01-elevationM must be finite.
PASS: P01-boreM must be finite.
PASS: P01-azimuthDeg must be finite.
PASS: P01-radialOffsetM must be finite.
PASS: P01-outsideDiameterM must be finite.
PASS: P01-projectionM must be finite.

PASS: P02-elevationM must be finite.
PASS: P02-boreM must be finite.
PASS: P02-azimuthDeg must be finite.
PASS: P02-radialOffsetM must be finite.
PASS: P02-outsideDiameterM must be finite.
PASS: P02-projectionM must be finite.
PASS: P03-elevationM must be finite.
PASS: P03-boreM must be finite.
PASS: P03-azimuthDeg must be finite.
PASS: P03-radialOffsetM must be finite.
PASS: P03-outsideDiameterM must be finite.
PASS: P03-projectionM must be finite.
PASS: P04-elevationM must be finite.
PASS: P04-boreM must be finite.
PASS: P04-azimuthDeg must be finite.
PASS: P04-radialOffsetM must be finite.
PASS: P04-outsideDiameterM must be finite.
PASS: P04-projectionM must be finite.
PASS: A01-elevationM must be finite.
PASS: A01-boreM must be finite.
PASS: A01-azimuthDeg must be finite.
PASS: A01-radialOffsetM must be finite.
PASS: A01-outsideDiameterM must be finite.
PASS: A01-projectionM must be finite.
PASS: A02-elevationM must be finite.
PASS: A02-boreM must be finite.
PASS: A02-azimuthDeg must be finite.
PASS: A02-radialOffsetM must be finite.
PASS: A02-outsideDiameterM must be finite.
PASS: A02-projectionM must be finite.
PASS: S01-elevationM must be finite.
PASS: S01-boreM must be finite.
PASS: S01-azimuthDeg must be finite.
PASS: S01-radialOffsetM must be finite.
PASS: S01-outsideDiameterM must be finite.
PASS: S01-projectionM must be finite.
PASS: S02-elevationM must be finite.
PASS: S02-boreM must be finite.
PASS: S02-azimuthDeg must be finite.
PASS: S02-radialOffsetM must be finite.
PASS: S02-outsideDiameterM must be finite.
PASS: S02-projectionM must be finite.
PASS: I01-elevationM must be finite.
PASS: I01-boreM must be finite.
PASS: I01-azimuthDeg must be finite.
PASS: I01-radialOffsetM must be finite.
PASS: I01-outsideDiameterM must be finite.
PASS: I01-projectionM must be finite.
PASS: I02-elevationM must be finite.
PASS: I02-boreM must be finite.
PASS: I02-azimuthDeg must be finite.
PASS: I02-radialOffsetM must be finite.

PASS: I02-outsideDiameterM must be finite.
PASS: I02-projectionM must be finite.
PASS: D01-elevationM must be finite.
PASS: D01-boreM must be finite.
PASS: D01-radialOffsetM must be finite.
PASS: D01-outsideDiameterM must be finite.
PASS: D01-projectionM must be finite.
PASS: D01-surfaceEdgeElevationMinM must be finite.
PASS: D01-surfaceEdgeElevationMaxM must be finite.
PASS: Entire neck cylinder fits ellipsoidal head.
PASS: V01-elevationM must be finite.
PASS: V01-boreM must be finite.
PASS: V01-azimuthDeg must be finite.
PASS: V01-radialOffsetM must be finite.
PASS: V01-outsideDiameterM must be finite.
PASS: V01-projectionM must be finite.
PASS: V01-surfaceEdgeElevationMinM must be finite.
PASS: V01-surfaceEdgeElevationMaxM must be finite.
PASS: compartment1-bottomM must be finite.
PASS: compartment1-topM must be finite.
PASS: compartment1-rotorM must be finite.
PASS: compartment2-bottomM must be finite.
PASS: compartment2-topM must be finite.
PASS: compartment2-rotorM must be finite.
PASS: compartment3-bottomM must be finite.
PASS: compartment3-topM must be finite.
PASS: compartment3-rotorM must be finite.
PASS: compartment4-bottomM must be finite.
PASS: compartment4-topM must be finite.
PASS: compartment4-rotorM must be finite.
PASS: compartment5-bottomM must be finite.
PASS: compartment5-topM must be finite.
PASS: compartment5-rotorM must be finite.
PASS: compartment6-bottomM must be finite.
PASS: compartment6-topM must be finite.
PASS: compartment6-rotorM must be finite.
PASS: compartment7-bottomM must be finite.
PASS: compartment7-topM must be finite.
PASS: compartment7-rotorM must be finite.
PASS: compartment8-bottomM must be finite.
PASS: compartment8-topM must be finite.
PASS: compartment8-rotorM must be finite.
PASS: compartment9-bottomM must be finite.
PASS: compartment9-topM must be finite.
PASS: compartment9-rotorM must be finite.
PASS: compartment10-bottomM must be finite.
PASS: compartment10-topM must be finite.
PASS: compartment10-rotorM must be finite.
PASS: compartment11-bottomM must be finite.
PASS: compartment11-topM must be finite.
PASS: compartment11-rotorM must be finite.
PASS: compartment12-bottomM must be finite.

PASS: compartment12-topM must be finite.
PASS: compartment12-rotorM must be finite.
PASS: compartment13-bottomM must be finite.
PASS: compartment13-topM must be finite.
PASS: compartment13-rotorM must be finite.
PASS: compartment14-bottomM must be finite.
PASS: compartment14-topM must be finite.
PASS: compartment14-rotorM must be finite.
PASS: compartment15-bottomM must be finite.
PASS: compartment15-topM must be finite.
PASS: compartment15-rotorM must be finite.
PASS: compartment16-bottomM must be finite.
PASS: compartment16-topM must be finite.
PASS: compartment16-rotorM must be finite.
PASS: compartment17-bottomM must be finite.
PASS: compartment17-topM must be finite.
PASS: compartment17-rotorM must be finite.
PASS: compartment18-bottomM must be finite.
PASS: compartment18-topM must be finite.
PASS: compartment18-rotorM must be finite.
PASS: compartment19-bottomM must be finite.
PASS: compartment19-topM must be finite.
PASS: compartment19-rotorM must be finite.
PASS: compartment20-bottomM must be finite.
PASS: compartment20-topM must be finite.
PASS: compartment20-rotorM must be finite.
PASS: compartment21-bottomM must be finite.
PASS: compartment21-topM must be finite.
PASS: compartment21-rotorM must be finite.
PASS: compartment22-bottomM must be finite.
PASS: compartment22-topM must be finite.
PASS: compartment22-rotorM must be finite.
PASS: compartment23-bottomM must be finite.
PASS: compartment23-topM must be finite.
PASS: compartment23-rotorM must be finite.
PASS: compartment24-bottomM must be finite.
PASS: compartment24-topM must be finite.
PASS: compartment24-rotorM must be finite.
PASS: compartment25-bottomM must be finite.
PASS: compartment25-topM must be finite.
PASS: compartment25-rotorM must be finite.
PASS: compartment26-bottomM must be finite.
PASS: compartment26-topM must be finite.
PASS: compartment26-rotorM must be finite.
PASS: compartment27-bottomM must be finite.
PASS: compartment27-topM must be finite.
PASS: compartment27-rotorM must be finite.
PASS: compartment28-bottomM must be finite.
PASS: compartment28-topM must be finite.
PASS: compartment28-rotorM must be finite.
PASS: compartment29-bottomM must be finite.
PASS: compartment29-topM must be finite.

PASS: Complete outward neck envelopes are disjoint (not merely bore openings).

PASS: Vertical vent cylinder clears full drive/pedestal/seal radial envelopes.

PASS: Actual ellipsoid/neck intersection envelopes clear internal supports and shaft lower termination.

PASS: Head connections and straight-shell stubs are axially disjoint.

ASSUMPTION: Class A: approved frozen Stage-3/4; no hydraulic/sizing recalculation.

ASSUMPTION: Class B: Perry 8e p15-83 / Pratt-Stevens design envelope; Garthe Table A.1 and Weber-Jupke 2020 DOI 10.1002/aic.16286 construction family.

ASSUMPTION: Class C: approved R1 proportions and stepped approximation, not exact reference reproduction or experimental validation.

ASSUMPTION: End zones are packaging allocations, not separation-performance sizing. Connection bores are geometric allowances, not hydraulic or DN sizing.

ASSUMPTION: NMP-down/RRBO-up is an explicit layout selection, not inferred from continuity. Vent is crown-region, not absolute-high-point qualified.

ASSUMPTION: N+1 plates follows selected shared-boundary and terminal-ring topology, not a universal source count.

OUTSIDE ENGINEERING SCOPE: Pressure-vessel shell/head thickness and code design

OUTSIDE ENGINEERING SCOPE: Shaft strength, deflection, critical speed

OUTSIDE ENGINEERING SCOPE: Bearing/seal ratings, drive power/torque and support structural qualification

OUTSIDE ENGINEERING SCOPE: Welds, keys, fasteners, flanges, reinforcement and external pipe routing

OUTSIDE ENGINEERING SCOPE: Separation residence-time, connection hydraulic capacity and operational venting qualification

Engineering notes: ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; system-generated; no normal-user construction inputs.

Revision notes: Browser verification fixture

Stage 3 result: 3; Stage 4 result: 4

Source SHA-256: hash-current

PRELIMINARY — NOT FOR FABRICATION. Mechanical verification, pressure-vessel design, shaft strength/deflection/critical speed, bearing/seal selection, drive adequacy and structural supports are outside this package.